EX NO:2 Chebyshev IIR Filter

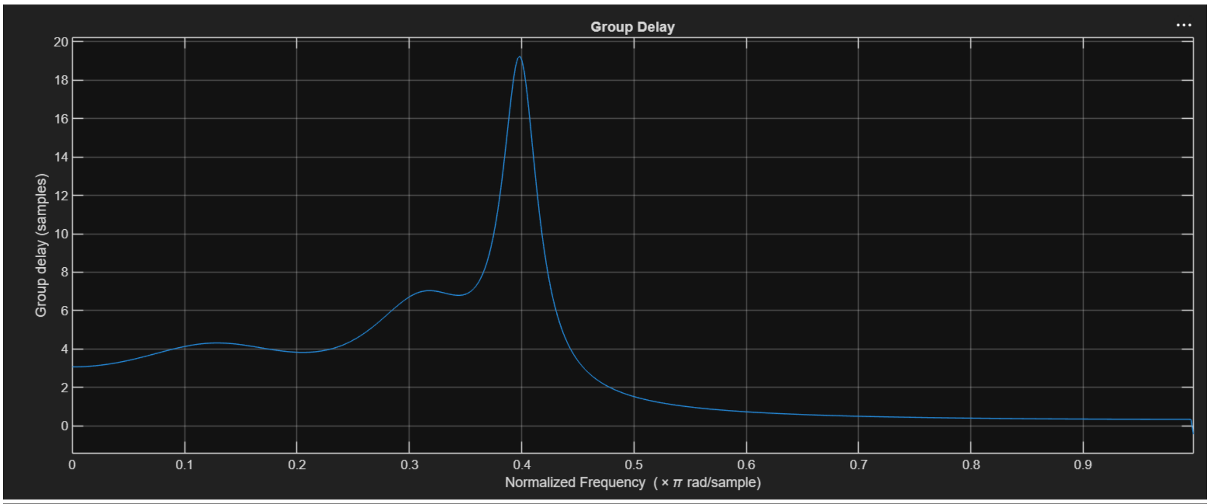
2.1. Design and Performance Evaluation of a Chebyshev Type-I IIR Low-Pass Filter using MATLAB

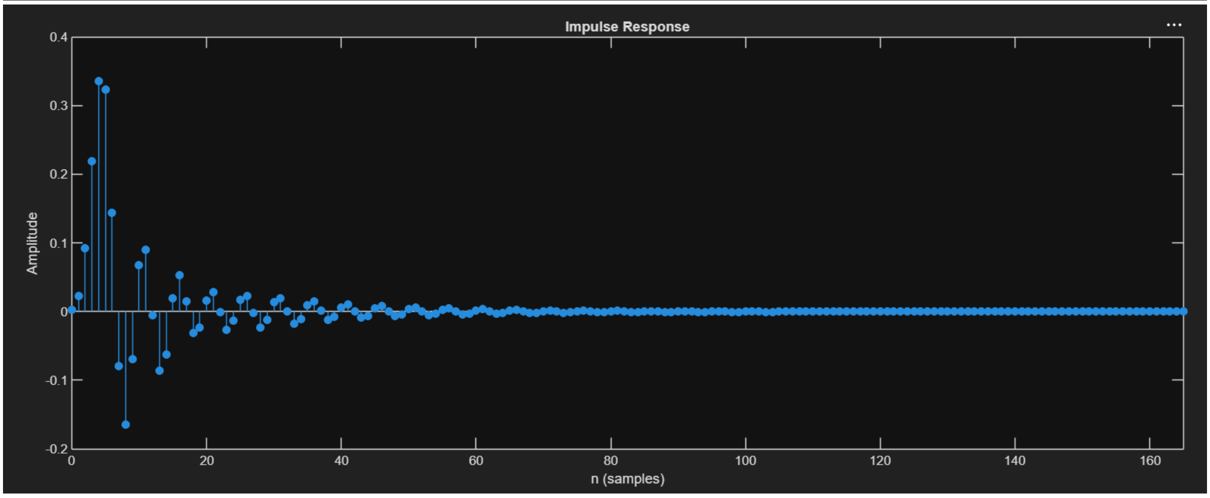
PROGRAM

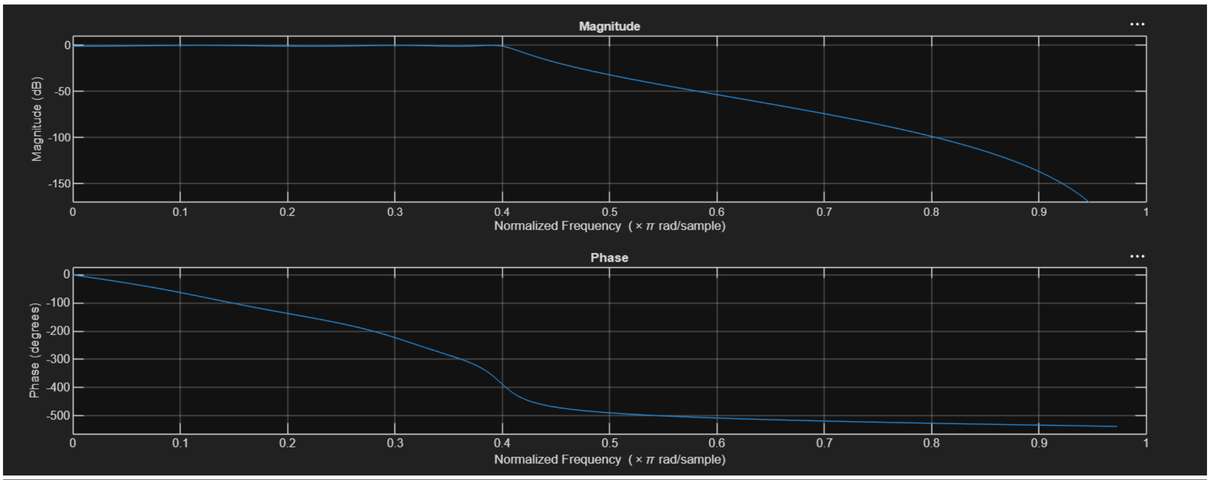
```
clc; clear;

Wp=0.4; Ws=0.55; Rp=1; As=40;

[N,Wn]=cheb1ord(Wp,Ws,Rp,As);

[b,a]=cheby1(N,Rp,Wn);

fprintf('Filter Order = %d\n',N);

fprintf('Cutoff Frequency = %.2f * pi rad/sample\n',Wn);

if all(abs(roots(a))<1)

   disp('System is STABLE');

else

   disp('System is UNSTABLE');

end

% -------- GRAPHS --------

figure; freqz(b,a);

% Magnitude + Phase

figure; impz(b,a);              % Impulse Response

figure; grpdelay(b,a);

% Group Delay

figure; zplane(b,a);            % Pole-Zero
```

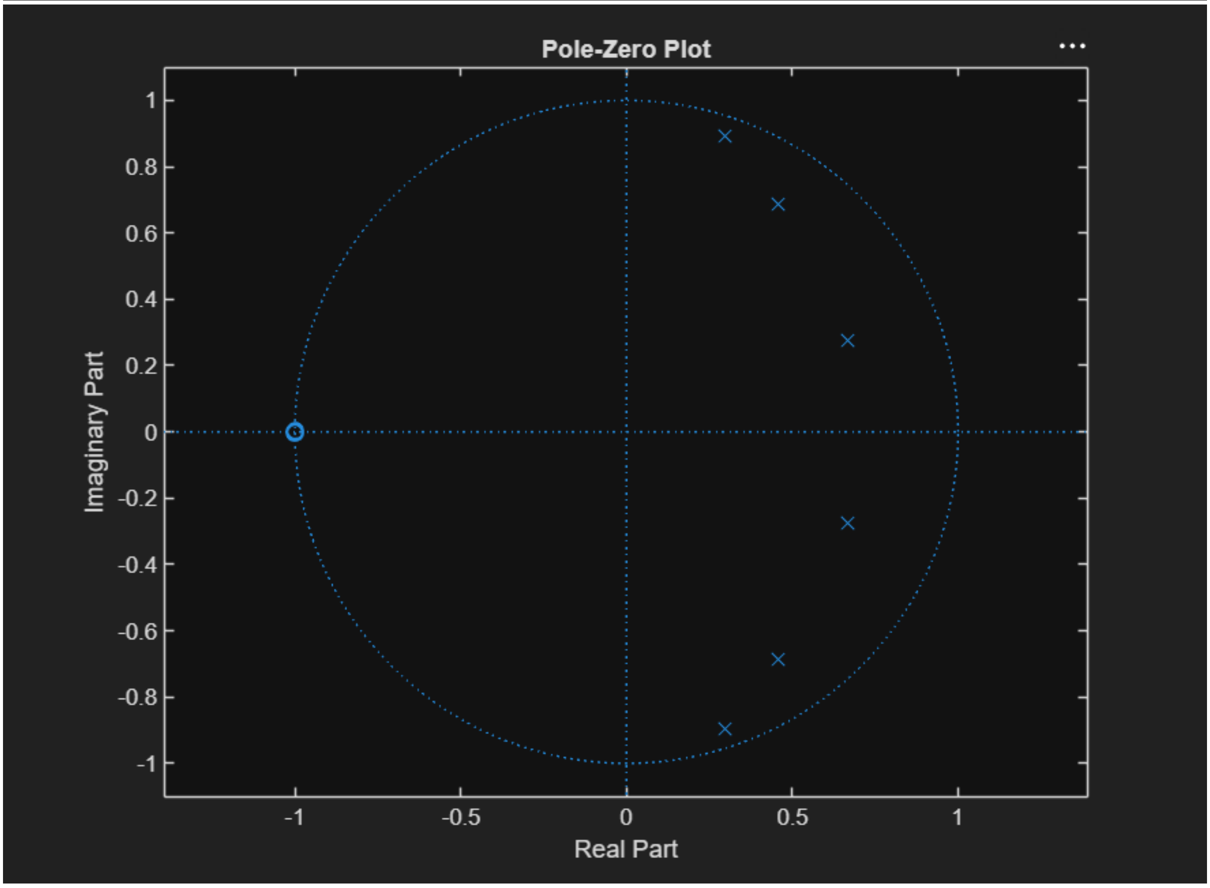
OUTPUT

Filter Order = 6
Cutoff Frequency = 0.40 * pi rad/sample
System is STABLE


Group Delay
Group delay (samples)
Normalized Frequency (× π rad/sample)
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
2
4
6
8
10
12
14
16
18
20


Impulse Response
Amplitude
n (samples)
0
20
40
60
80
100
120
140
160
-0.2
-0.1
0.1
0.2
0.3
0.4


Magnitude
Magnitude (dB)
0
-50
-100
-150
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Normalized Frequency (×π rad/sample)
Phase
Phase (degrees)
0
-100
-200
-300
-400
-500
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Normalized Frequency (×π rad/sample)


Pole-Zero Plot
Imaginary Part
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-1
-0.5
0
0.5
1
Real Part